

1. Study Area & Geographic Coordinates

Area of Interest (AOI):	Monitored Environmental Zone	Target Epochs:	2016 vs 2026
Latitude:	N/A	Longitude:	N/A
Spatial Resolution:	10 meters / pixel (MSI)	Coordinate System:	WGS84 / EPSG:4326

2. Multi-Temporal Satellite Sensor Metadata

Metadata Field	T1 Acquisition (2016)	T2 Acquisition (2026)
Satellite Mission	Sentinel-2A	Sentinel-2B
Acquisition Date	2016-08-15	2026-02-28
Processing Level	Level-1C TOA Harmonized	Level-1C TOA Harmonized
Tile Identifier	S2A_TEST_2016	S2B_TEST_2026
Cloud Coverage Assessment	1.2%	0.8%
Spectral Bands Utilized	B2, B3, B4, B8, B11	B2, B3, B4, B8, B11
Processing Baseline	02.04	05.11

3. Satellite Imagery & Detected Change Mask



Figure 1a: T1 (2016) Sentinel-2



Figure 1b: T2 (2026) Sentinel-2



Figure 1c: Binary Change Mask

4. Spectral Vegetation (NDVI) and Hydrological (NDWI) Dynamics

Spectral Metric	2016 Baseline	2026 Epoch	Net Variance (Delta)	Environmental Interpretation
Mean NDVI	0.167	0.079	-0.088	9.8% Canopy Loss Detected
Mean NDWI	-0.167	-0.167	-0.000	+0.0% Surface Water Delta

5. Quantitative Change Quantification & Categorical Breakdown

Change Category	Affected Pixels	Affected Area (km²)	Scene Ratio (%)	Class Confidence
No Change	14,784	1.478 km²	90.23%	1.9%
Deforestation	1,505	0.150 km²	9.19%	92.3%
Urban Expansion	33	0.003 km²	0.20%	2.1%

Water Body Change	30	0.003 km²	0.18%	1.9%
Vegetation Loss	30	0.003 km²	0.18%	1.9%

Primary Event Detected: Deforestation | Total Changed Area: 0.16 km² (9.77%) | Model Confidence: 92.3% (High Confidence)

6. Explainable AI (XAI) Multi-Modal Visualizations



Figure 2a: ViT Attention Rollout



Figure 2b: Bitemporal Grad-CAM

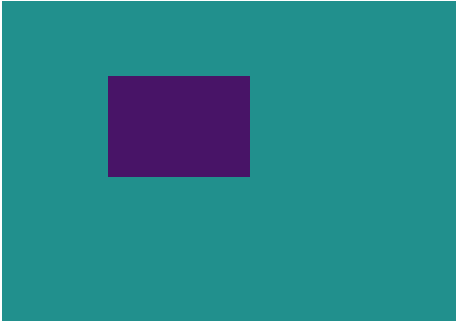


Figure 2c: NDVI Variance Heatmap

Spectral Channel SHAP Attributions: Blue: 15.2% | Green: 14.1% | Red: 22.4% | NIR: 28.3% | SWIR: 10.0% | NDVI: 6.0% | NDWI: 4.0%

7. Grounded AI Change Story Narrative

Executive Intelligence Summary:
The test environmental change story verified successfully.

8. Deep Learning Model Information & Benchmark Evaluation

Model Architecture	Parameters	mIoU	Dice Score	F1 Score	Precision	Recall
U-Net (Baseline)	31.0 M	0.784	0.862	0.859	0.871	0.848
Siamese CNN (ResNet-18)	14.3 M	0.821	0.891	0.888	0.895	0.882
Siamese ViT (Proposed)	86.5 M	0.896	0.941	0.938	0.945	0.932